

Advanced Machine Learning

Assignment 2

Manolescu Cosmin Mihai

June 2020

1. Consider $H = \{h_{\theta_1} : \mathbb{R} \rightarrow \{0, 1\}, h_{\theta_1}(x) = 1_{[x \geq \theta_1]}(x) = 1_{[\theta_1, +\infty)}(x), \theta_1 \in \mathbb{R}\} \cup \{h_{\theta_2}(x) = 1_{[x < \theta_2]}(x) = 1_{(-\infty, \theta_2)}(x), \theta_2 \in \mathbb{R}\}$

- a.** Compute the shattering coefficient $\tau_H(m)$ of the growth function for $m \geq 0$.
 The growth function that we need to compute for H , $\tau_H(m)$ is associated with the number of different functions from a set C of size m to $\{0, 1\}$ that can be obtained by restricting H to C . In other words, we need to find the general form for how many functions can correctly classify a set S for $m \geq 0$.

We know that if $VCdim(\mathbb{H}) = d$ then for all $m \leq d$ we have $\tau_H(m) = 2^m$. We need to compute the $VCdim(H)$ in order to find the point where the function changes from an exponential to a polynomial form (stated by Sauer's Lemma).

For $m = 1$ we will have a single point on the $\mathbb{R}$ axis. It is clear that we can choose either a point θ_1 such that $\theta_1 < x$ or θ_2 such that $\theta_2 > x$. From this it results that $VCdim(H) \geq 1$.

For $m = 2$ we will have two ordered numbers x_1 and x_2 on the $\mathbb{R}$ axis, as in the following figure.



Figure 1: Ordered points in space

For those points we have 4 possible labels combinations: $(0, 0), (1, 0), (1, 1), (0, 1)$.

- For the point $(0, 0)$ we will take a hypothesis H with $\theta_1 > x_2 > x_1$ or a hypothesis H with $\theta_2 < x_1 < x_2$.
- For the point $(1, 1)$ we will take a hypothesis H with $\theta_1 \leq x_1 \leq x_2$ or a hypothesis H with $\theta_2 > x_2 \geq x_1$.

- For the point $(0, 1)$ we will take a hypothesis H with $x_2 \geq \theta_1 > x_1$.
- For the point $(1, 0)$ we will take a hypothesis H with $x_1 \leq \theta_2 < x_2$.

We can conclude that $VCdim(H) \geq 2$. **(1)**

For $m = 3$ We cannot classify for example a sequence of labels of alternating labels, like $(0, 1, 0)$. Thus, $VCdim(H) < 3$. **(2)**

From **(1)** and **(2)** $\implies VCdim(H) = 2$.

For $m \leq 2$ we will have $\tau_H(m) = 2^m$.

We need to see now what is $\tau_H(m)$ for $m \geq 3$.

We saw earlier that we cannot get a hypothesis that can correctly classify the alternating labels. However, we can find a classifier that correctly classifies sequence, non-alternating labels. For example, for $m = 3$ we have the following labels that can be correctly classified: $(0,0,0)$, $(0,0,1)$, $(0,1,1)$, $(1,1,1)$, $(1, 1, 0)$ and $(1,0,0)$. For any m we have $2m$ such labels sequences. Thus, we can conclude that $\tau_H(m) = 2m$, for $m \geq 3$. We can define τ by the following formula:

$$\tau_H(x) = \begin{cases} 2^m & \text{if } m \leq 2, \\ 2m & \text{if } m \geq 3. \end{cases}$$

- b.** Compare your result with the general upper bound for the growth functions.

The general upper bound for the growth function is computed using the Sauer's Lemma as it follows:

$$\tau_H(m) \leq \sum_{i=0}^d C_m^i = \sum_{i=0}^2 \frac{m!}{i!(m-i)!} = \frac{m^2 + m + 2}{2}$$

It is clear that for $m \leq 2$, $\tau_H(m) \leq \frac{m^2+m+2}{2}$. We need to check that the inequality is verified for $m \geq 3$.

$$2m \leq \frac{m^2 + m + 2}{2} \iff m^2 - 3m + 2 \geq 0$$

The roots of this function are 1 and 2. For ≥ 2 the function $m^2 - 3m + 2 \geq 0$ due to the fact that this function is monotonically increasing for values after 2 $\Rightarrow \tau_H(m)$ is upper bounded by the general upper bound.

- c.** Does there exist a hypothesis class $\mathbb{H}$ for which $\tau_H(m)$ is equal to the general upper bound (over $\mathbb{R}$ or another domain $\mathbb{X}$)? If your answer is yes please provide an example, if your answer is no please provide a justification.

We saw that for the previous hypothesis we obtained a VC-dimension of 2 and a growth function that was upper bounded by $\frac{m^2+m+2}{2}$. In order

to have the growth function equal with the general upper bound we can give two examples of hypothesis that satisfy this request. Those two are

$$H = \{h_\theta : \mathbb{R} \rightarrow \{0, 1\}, h_\theta(x) = 1_{[x \geq \theta]}(x) = 1_{[\theta, +\infty)}(x), \theta \in \mathbb{R}\}$$

and

$$H = \{h_\theta : \mathbb{R} \rightarrow \{0, 1\}, h_\theta(x) = 1_{[x < \theta]}(x) = 1_{(-\infty, \theta)}(x), \theta \in \mathbb{R}\}$$

We can see that those hypothesis are the ones that are reunited for the first two subtasks. We will demonstrate the equality for the first hypothesis only, for the second one the demonstration being analogue. We can see that for the first hypothesis the VC-dimension is 1, because for $m = 2$ we cannot find θ such that we can correctly classify $(1, 0)$. For $m = 1$ we can find a θ in order to correctly classify one point. For $VCdim(H) = 1 \Rightarrow \tau_H(m) \leq \sum_{i=1}^1 C_m^i \iff \tau_H(m) \leq \frac{m!}{1!(m-1)!} \iff \tau_H(m) \leq m$.

The general formula for $\tau_H(m)$ for $m > 1$ will be the general formula for the first two tasks divided by two, because we can only take the sequences of labels that start with 0 and end with 1. From this we can conclude that $\tau_H(m) = m$ which is the general upper bound.

The same logic applies for the second hypothesis, only that for this one θ will be chosen to be higher than the positive examples.

2. Let Σ be a finite alphabet and let $\mathbb{X} = \Sigma^m$ be a sample space of all strings of length m over σ . Let $\mathbb{H}$ be a hypothesis space over $\mathbb{X}$, where $\mathbb{H} = \{h_w : \Sigma^m \rightarrow \{0, 1\}, w \in \Sigma^*, 0 < |w| \leq m, s.t. h_w(x) = 1 \text{ if } w \text{ is a substring of } x\}$.

- a. Give an upper bound (any upper bound that you can come up with) of the VC-dimension of $\mathbb{H}$ in terms of $|\Sigma|$ and m . In order to find an upper bound for the VC-dimension for H we will use the following upper bound:

$$VCdim(H) \leq \log_2 |H|$$

The dimension of possibilities for H is defined by all the possible labels for a training set of size n and by all the possible strings generated from the alphabet Σ with size m . All the possible strings that can be generated are $|\Sigma|^m$. There are also 2^n possible labels combinations. From those two, we can conclude that

$$|H| = 2^n |\Sigma|^m \Rightarrow VCdim(H) \leq \log_2 2^n |\Sigma|^m$$

- b. Give an efficient algorithm for finding a hypothesis h_w consistent with a training set in the realizable case. What is the complexity of your algorithm?

Example: let $\Sigma = \{a, b, c\}$, $m = 4$ and the training set $S = \{(abc, 1)(baca, 0), (bcac, 0), (abba, 1)\}$. The output of the algorithm should be h_{ab} .

1. For the alphabet Σ build all the possible $\frac{m(m+1)}{2}$ substrings. We will name this list of string Σ_s . Knowing that we're in the realizable case, one of the substrings from this list will be the substring for our hypothesis.
2. For each examples in S , for all negative examples, remove from Σ_s the substrings that are in those and for the positive examples, remove the substrings that are not in those strings.

The list Σ_s will have left the substring that can be used to build a hypothesis h_{Σ_s} with the remaining substring, knowing that this substring will be a substring for the positive examples and will not be a substring for the negative ones. The complexity for the first step is $O(m^2)$. For the second step the time complexity will be $O(|S|m^2)$. The overall final time complexity is

$$O(m^2 + |S|m^2) = O(m^2(|S| + 1)) = O(m^2|S|)$$

3. Consider the boosting algorithm described (page 4) in the article "[Rapid object detection using a boosted cascade of simple features](#)", P. Viola and M. Jones, CVPR 2001. Consider that the number of positives is equal with the number of negative examples ($l=m$).

- a. Prove that the distribution w_{t+1} obtained at round $t + 1$ based on the algorithm described in the article is the same with the distribution $\mathbf{D}^{(t+1)}$ obtained on the procedure described in lecture 11 (slides 10-12).

$$D^{(t+1)}(i) = \frac{D^{(t)}(i)e^{-w_t h_t(x_i)y_i}}{\sum_{i=1}^m D^{(t)}(i)e^{-w_t h_t(x_i)y_i}},$$

where $w_t = \frac{1}{2} \ln\left(\frac{1}{\epsilon_t} - 1\right)$

This update function for the weights distribution can be rewritten in the following form:

$$D^{(t+1)}(i) = \begin{cases} D^{(t)}(i) \frac{\sqrt{\frac{\epsilon_t}{1-\epsilon_t}}}{\sum_{i=1}^m D^{(t)}(i) e^{-w_t h_t(x_i)y_i}} & \text{if we missclassify,} \\ D^{(t)}(i) \frac{\sqrt{\frac{1-\epsilon_t}{\epsilon_t}}}{\sum_{i=1}^m D^{(t)}(i) e^{-w_t h_t(x_i)y_i}} & \text{otherwise.} \end{cases}$$

We want to prove that this update function is equivalent with the one for $w_{t+1,i}$ from the paper.

$$w_{t+1,i} = w_{t,i} \frac{\left(\frac{\epsilon_t}{1-\epsilon_t}\right)^{1-e_i}}{\sum_{i=1}^m w_{t,i}},$$

where $e_i = 1$ if we missclassify and 0 otherwise.

If the sum from the denominator $\sum_{i=1}^m D^{(t)}(i) e^{-w_t h_t(x_i)y_i}$ is equal to $\sum_{i=1}^m D^{(t)}(i) \sqrt{\frac{1-\epsilon_t}{\epsilon_t}}$ then we will have the following cases:

1. For missclassification we will have

$$D^{(t+1)}(i) = \frac{D^{(t)}(i) \sqrt{\frac{\epsilon_t}{1-\epsilon_t}}}{\sum_{i=1}^m D^{(t)}(i) \sqrt{\frac{1-\epsilon_t}{\epsilon_t}}} = \frac{D^{(t)}(i) \frac{\epsilon_t}{1-\epsilon_t}}{\sum_{i=1}^m D^{(t)}(i)}$$

2. For correct classification we will have

$$D^{(t+1)}(i) = \frac{D^{(t)}(i) \sqrt{\frac{1-\epsilon_t}{\epsilon_t}}}{\sum_{i=1}^m D^{(t)}(i) \sqrt{\frac{1-\epsilon_t}{\epsilon_t}}} = \frac{D^{(t)}(i)}{\sum_{i=1}^m D^{(t)}(i)}$$

We can write this final update function as it follows

$$D^{(t+1)}(i) = \begin{cases} \frac{D^{(t)}(i) \frac{\epsilon_t}{1-\epsilon_t}}{\sum_{i=1}^m D^{(t)}(i)} & \text{if we missclassify,} \\ \frac{D^{(t)}(i)}{\sum_{i=1}^m D^{(t)}(i)} & \text{otherwise.} \end{cases}$$

If we take a closer look to the update function from the paper we can rewrite it in the following form

$$w_{t+1,i} = \begin{cases} \frac{w_{t,i} \frac{\epsilon_t}{1-\epsilon_t}}{\sum_{i=1}^m w_{t,i}} & \text{if we missclassify,} \\ \frac{w_{t,i}}{\sum_{i=1}^m w_{t,i}} & \text{otherwise.} \end{cases}$$

We can clearly see now that those two update functions are actually equivalent.

- b. Prove that the two final classifiers (the one described in the article and the one described in the lecture) are equivalent.

The final classifier from the paper, with $y = 0, 1$ is the following

$$h(x) = \begin{cases} 1 & \text{if } \sum_{t=1}^T \alpha_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^T T \alpha_t, \\ 0 & \text{otherwise.} \end{cases}$$

We want to prove that the following classifier, with $y = \{-1, 1\}$, is equivalent with the one above:

$$\text{sign}\left(\sum_{t=1}^T w_t h_t(x)\right)$$

The form from above can be rewritten in the following way:

$$h(x) = \begin{cases} 1 & \text{if } \sum_{t=1}^T w_t h_t(x) > 0, \\ 0 & \text{otherwise.} \end{cases}$$

From this equation we can see that in order to give a positive prediction, the weak classifiers that predict true times their weights have to sum up to a higher value than the weak classifiers times their weights. We can write then the following equation:

$$\sum w_t^+ - \sum w_t^- > 0,$$

where w_t^+ are the weights for the classifiers that predicted positive and w_t^- are the weights for the classifiers that predicted negative. The main idea of this sum in order to predict positive is that

$$\sum w_t^+ > \sum w_t^- \iff \sum w_t^+ > \frac{1}{2} \sum_{t=1}^T w_t \quad (1)$$

Let's assume that $\sum_{t=1}^T w_t h_t(x) = 0$. In this case it would mean that $\sum w_t^+ = \sum w_t^- = \frac{1}{2} \sum_{t=1}^T w_t$. If $\sum_{t=1}^T w_t h_t(x) > 0 \Rightarrow \sum w_t^+ > \sum w_t^- > \frac{1}{2} \sum_{t=1}^T w_t$. We can apply the same logic for $\sum_{t=1}^T w_t h_t(x) < 0$. Using those arguments, equation (1) is true.

If we change the output space from $y = \{-1, 1\}$ to $y = \{0, 1\}$ we can rewrite the classifier from the course as it follows:

$$h(x) = \begin{cases} 1 & \text{if } \sum_{t=1}^T w_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^T w_t, \\ 0 & \text{otherwise.} \end{cases}$$

which is equivalent with the classifier from the paper.